

# Kareem Abdelmaqsoud

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## INTERESTS

AI for Science, Material Science, Catalysis, Machine Learning, Large Language Models (LLMs), and Data Science

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## EDUCATION

### Carnegie Mellon University

PhD in Chemical Engineering – GPA 3.96

Pittsburgh, PA  
August 2022 – December 2026

Research Advisors: John R. Kitchin & Andrew J. Gellman

Selected Coursework: Data science & ML – Deep learning – AI Agents

Industry collaborations: Meta Fundamental AI Research (Fairchem) team

### University of Rochester

Bachelor of Science - Chemical Engineering – GPA 3.95

Rochester, NY  
August 2018 - May 2022

Research Advisor: Andrew D. White

Selected Coursework: ML for Molecules and Materials

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## CURRENT RESEARCH PROJECTS

### Agentic Workflow for Improving Electrochemical Interfaces

Carnegie Mellon University

February 2026 – current

- Built an LLM agentic workflow for autonomous surface-doping optimization of water–solid electrochemical interfaces using molecular dynamics (MD).
- Integrated LangGraph, literature search, LAMMPS simulations, and ChromaDB database, significantly improving hydrogen bonding in 4 of 9 tested systems.

### Computational Design of Rare-earth Metal Alloys

Carnegie Mellon University

August 2024 – current

- Built an ML-guided materials discovery pipeline reducing experimental search space by 10× and enabling discovery of a ductile tungsten alloy validated experimentally.
  - Developed interpretable physics-based model linking electronic structure features to alloy ductility, enabling predictive screening of new alloys.
  - Built automated high-throughput materials simulation pipelines integrating DFT, ML potentials, and CALPHAD for phase diagram prediction.
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## INDUSTRY EXPERIENCE

### Bosch USA

Watertown, MA

Computational Materials Science Intern

May – August 2026

- Applying universal ML potentials to build a phase diagram to model Aluminum Germanium (AlGe) alloy for semiconductor applications in MEMS devices
- Study the effect of Si contamination on phase stability and eutectic composition of the AlGe alloy.

### Orbital Materials

Princeton, NJ

Machine Learning Researcher Intern

May – August 2025

- Within a 3-month internship, I developed OrbMol, a large-scale ML potential trained on 100M organic molecular structures incorporating magnetism and charge effects.
- Achieved state-of-the-art accuracy compared with existing molecular ML potentials.
- Contributed to model architecture, large-scale dataset preparation, and training pipelines.

### Entos (Now Iambic Therapeutics)

Remote

Machine Learning Engineer Intern

January – July 2022

- Performed hyperparameter tuning for the model used by the experimental drug discovery team to improve its accuracy
- Implemented new pretraining methods such as contrastive learning that led to 20% improvement in accuracy

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## SKILLS

- **Programming:** Python
  - **Frameworks:** Kubernetes, AWS, Git, MongoDB, atomate, FastMCP, and LangChain
  - **Deep Learning & Data science:** PyTorch, TensorFlow, NumPy, Pandas, and scikit-learn
  - **Packages:** ASE, Pymatgen, VASP for DFT, and CALPHAD, LAMMPS for molecular dynamics
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## HONORS & AWARDS

- Richard King Mellon Foundation Presidential Fellowship in Energy.
  - 1<sup>st</sup> Place Award in the Undergraduate Poster Competition at the AIChE conference, Boston, MA.
  - Research & Innovation Grant Recipient for my undergraduate research.
  - Tau Beta Pi Engineering Honor Society.
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## PUBLICATIONS

### **Investigating the Error Imbalance of Large-Scale Machine Learning Potentials in Catalysis (Catalysis Science & Technology)**

Abdelmaqsoud, K., Shuaibi, M., Kolluru, A., Cheula, R., & Kitchin, J. R.

### **Structure-Sensitive Reaction Kinetics of Chiral Molecules on Intrinsically Chiral Surfaces (Journal of Phys Chem)**

Abdelmaqsoud, K., Radetic, M., Fernandez-Caban, C., Widom, M., Kitchin, J. R., & Gellman, A. J.

### **Uncertainty Quantification in Graph Neural Networks with Shallow Ensembles (MLST)**

Vinchurkar, T., Abdelmaqsoud, K., & Kitchin, J. R.

### **Computational Design of Ductile Additively Manufactured Tungsten-Based Refractory Alloys (Comp Mat Science)**

Abdelmaqsoud, K., Sinclair, D., Karra, V. S. S. A., Taheri-Mousavi, S. M., Widom, M., Webler, B. A., & Kitchin, J. R.

### **Electronic Structure and Elasticity of the Ta–W Solid Solution (Physical Rev Mat)**

Abdelmaqsoud, K., Kitchin, J. R., & Widom, M.

### **UMA: A Family of Universal Models for Atoms (NeurIPS)**

Wood, B. M., Dzamba, M., Fu, X., Gao, M., Shuaibi, M., Barroso-Luque, L., Abdelmaqsoud, K., Gharakhanyan, V., Kitchin, J. R., Levine, D. S., Michel, K., Sriram, A., Cohen, T., Das, A., Rizvi, A., Sahoo, S. J., Ulissi, Z. W., & Zitnick, C. L.

### **The Open DAC 2025 Dataset for Sorbent Discovery in Direct Air Capture (Nature Chem Eng)**

Sriram, A., Brabson, L. M., Yu, X., Choi, S., Abdelmaqsoud, K., Moubarak, E., de Haan, P., Löwe, S., Brehmer, J., Kitchin, J. R., Welling, M., Zitnick, C. L., Ulissi, Z. W., Medford, A. J., & Sholl, D. S.